MIL-M-80173B
21 November 1986
SUPERSEDING
MIL-M-80173A
25 June 1982

MILITARY SPECIFICATION

MANIPULATORS, AUTOMATIC WELDING HEAD, STATIONARY BASE AND TRAVELING BOOM, ELECTRIC MOTOR DRIVEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric motor driven manipulators with stationary bases and traveling booms for use with automatic welding heads. Welding heads are not covered by this specification.

1.2 Classification. Manipulators covered by this specification are of one type and shall be of the sizes listed in table I. The size to be furnished shall be as specified (see 6.2.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, TN 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferrred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Suseptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government publications. The following other Government publication forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR, Chapter XVII, Part 1910 - Occupational Safety and Health Standards.

(Copies of specifications, standards, and other Government publications required by contrators in connection with specific acquistion functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongoverment document which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 1 - Industrial Controls and Systems.

NEMA MG 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association 2101 L Street, N.W. Washington, DC 20037-1570.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standards for Industrial Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The manipulator shall be new and one of the manufacturer's current models capable of transporting an attached automatic welding head and associated equipment along a required path for horizontal and circumferential welding in accordance with the requirements herein. The manipulator shall include all components, parts, and features necessary to meet the performance requirements specified herein. Additional features which are not specified herein, but are a part of the manufacturer's current design, shall be included with the manipulator. The manipulator shall have a stationary base and a column with 360 degree power operated rotary motion of the column, and a ram or boom with lift and reach travel. The manipulator shall be designed to operate satisfactorily in ambient temperatures up to 112 degrees Fahrenheit. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Dimensional systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the manipulator. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all dimensions, measurements, sizes, and capacities are given in US units. These dimensions may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particu-

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lar graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the manipulator shall be graduated in the same system.

3.2.2 Reclaimed materials. The manipulator may contain reclaimed materials provided such materials will not jeopardize the intended use and performance of the manipulator. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the manipulator.

3.2.3 Energy efficiency. The manipulator and its components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located conveniently to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the manipulator that present safety hazards. The safety devices shall not interfere with the operation of the manipulator. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All manipulator parts, components, mechanisms, and assemblies furnished on the manipulator, whether or not specifically required herein, shall comply with all of the requirements contained in 29 CFR, Chapter XVII, Part 1910. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.6 Mercury restriction. The manipulator shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the manipulator.

3.2.8 Environmental protection. The manipulator shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the manipulator shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation (see 6.4).

3.2.9 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or manipulator.

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3.3 Construction. The manipulator shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the manipulator at its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the manipulator. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the manipulator shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the manipulator and its attachments and accessories shall conform to FED-STD-H28.

3.3.7 Dials. All dials that indicate manipulator lift travel, reach travel motion in inches per minute (ipm), or rotational motion in revolutions per minute (rpm) shall be such that graduations may be read from the normal operating position. Dials shall be permanently and legibly engraved or etched on a nonglare background.

3.3.8 Gears. All gears shall be constructed from a material suitable for the intended purpose. All drive gears shall have the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.3.9 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

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3.4 Components. The basic components of the manipulator shall include the base, column, crosshead, ram or boom, and controls. The base, column, crosshead, and ram or boom, shall be constructed of cast ferrous metals or welded structural shapes. Size and rigidity shall be adequate to meet the performance requirements specified herein. Roller motion shall be accomplished with adjustable and replaceable roller units. Slide motion shall be accomplished on hardened ways with adjustable and replaceable gib arrangement. Wipers shall be provided to remove dirt and other contaminants from the ways.

3.4.1 Stationary base. The stationary base shall function as a support for all components of the manipulator and shall provide 360 degree rotation of the column. Rotating motion of the column shall be accomplished with a reversible, infinitely variable speed silicon control rectifier (SCR) direct current (DC) motor drive. The rate of rotation shall be not less than 0.10 rpm. Provisions shall be made to lock the column at any rotary position. Unless otherwise specified, the base shall be of the free standing type and shall have radially extented legs for rigidity. Portability and leveling shall be accomplished by casters, locking devices, and jack screws. When specified (see 6.2.1), the base shall be of the bolt-down type with bolt holes for mounting the base to the floor.

3.4.2 Column. The column shall provide rotary and vertical motion of the crosshead. The column shall also support the control center with connections to the electrical power supplies and welding accessories. Vertical motion of the crosshead on the column shall be on precision way surfaces suitable to provide the required performance. Vertical motion of the crosshead shall be accomplished with a SCR electronic drive system. Vertical motion drive shall be the manufacturer's standard. All gears or screw drives shall provide self-locking, anti-fall capabilities and all chain drives shall include a ball type anti-fall device. All drives shall include either multiple units, each capable of supporting the full rated capacity load on the crosshead, or a single unit capable of supporting at least twice the full rated load on the crosshead. The vertical drive shall be mounted on the column or crosshead, and shall provide the lift required by table I.

3.4.3 Crosshead. The crosshead shall travel vertically on the column, and shall support horizontal motion of the ram or boom. The crosshead shall guide horizontal travel of the ram or boom through, across, or on the crosshead with reach conforming to the requirements of table I. Horizontal ram or boom travel shall be accomplished with a reversible, infinitely variable speed SCR electronic drive system and shall be of the manufacturer's standard design.

3.4.4 Ram or boom. The ram or boom shall travel horizontally from the crosshead and shall support the total load of the automatic welding head and accessories. The ram or boom lift and reach shall be as specified in table I.

3.4.4.1 Ram or boom lift rate of travel. The rate of ram or boom lift travel shall be constant throughout the lift travel range. The lift rate of

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travel shall range from a minimum of not more than .30 ipm to a maximum of not less than 27 ipm. Control of the lift rate of travel shall provide positive repeatability and positionability throughout the lift range specified, and shall deviate no more than a plus or minus 2 percent from the control set point.

3.4.4.2 Ram or boom reach rate of travel. The ram or boom reach rate of travel shall range from a minimum rate of not more than 6 ipm to a maximum rate of not less than 60 ipm. Control of reach travel rate shall provide positive repeatability and positionability throughout the reach range specified, and shall deviate not more than a plus or minus 2 percent from the control set point.

3.4.5 Accuracy. The rate of sliding motion shall be within plus or minus 1 percent of the set rate throughout the full lift and reach traveling ranges with any load up to the maximum design load. Unless otherwise specified (see 6.2.1), the path of travel deviation shall be within plus or minus .005 of an inch per linear foot to permit quality welding with the automatic submerged arc process on material of 3/8 to 1-1/2 inch in thickness and automatic precision Metal Inert Gas (MIG) Welding or Tungsten Inert Gas (TIG) Welding of materials ranging from 10 gauge to 1/2 inch in thickness.

3.4.6 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to NFPA 79. Each manipulator shall draw all its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit and shall have a fused safety disconnect switch or circuit breaker. The manipulator shall be initially wired for operation on 460 volts. An identified terminal for grounding the manipulator when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for disconnecting the size grounding conductor specified in NFPA 79 for the disconnect fuse rating. Electronic type drive controls shall be mounted in an enclosure conforming to NEMA ICS 1.

3.4.6.1 Motors. Unless otherwise specified (see 6.2.1), motors shall be rated for continuous duty, and shall be equipped with ball bearings. Bearings shall be of the sealed and permanently lubricated type, or the shielded and sealed type with a means for lubricant change. Motor characteristics shall conform to NEMA MG 1. Motor horsepower shall be of sufficient capacity to meet the specified performance. Motors shall be totally enclosed.

3.4.7 Controls. The controls for operating the manipulator shall be assembled in a pendant type enclosure with an attached multi-conductor cable not less than 20 feet long. The controls for the manipulator shall consist of all equipment required to operate the manipulator. The controls for the manipulator shall include a flashing red warning light located at the top of the mast and end of the boom and an emergency stop button to stop all functions of the manipulator. Unless otherwise specified (see 6.2.1), the controls shall include not less than the following:

a. Separate potentiometer type control for the reach, lift, and rotation, modes of motion.

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b. Potentiometer type control for variable maintained speeds.

c. Potentiometer type control for inching motion.

d. Forward and reverse selector switch.

e. Emergency stop switch.

3.4.8 Solid state components. Solid state design shall be used throughout for electronic components.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the manipulators shall not be less than the requirements specified in table I for the manipulator size specified.

3.5.1 Load capacity. The load capacity for size 1 and size 2 manipulators shall be not less than 300 pounds and for size 3 and larger units not less than 500 pounds equally distributed on both ends of the ram or boom. The load capacity requirements shall be at the maximum lift and maximum reach for all sizes.

3.6 Performance. The manipulators shall be capable of moving an automatic welding head and associated equipment along the reach and lift paths at the rates of motion specified herein. Motion shall be smooth with no jerking or oscillating action which will affect welding performance.

3.7 Standard equipment. Unless otherwise specified, all equipment normally furnished by the manufacturer on his standard commercial manipulator shall be furnished. When specified (see 6.2.1), required additional equipment shall be furnished and fully described.

3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.8.1 Welding head mount. The mounting adaptors for the welding head shall support the head in a position for both horizontal and vertical welding, with adjustments for welding at angles between horizontal and vertical.

3.8.2 Operator's fixtures. Fixtures such as operator's ladder, walk way, and seat shall be furnished with specific characteristics as specified.

3.8.3 Indicating meters. Indicating meters shall be calibrated in a range to indicate the full range of motion measured.

3.9 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.10 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

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TABLE I. Sizes and capacities.

Size	Ram or boom lift travel (feet)	Ram or boom reach travel (feet)	Size	Ram or boom lift travel (feet)	Ram or boom reach travel (feet)
1	4	4	19	12	15
2	5	5	20	12	16
3	6	6	21	14	11
4	6	7	22	14	14
5	6	8	23	14	15
6	4	10	24	15	12
7	8	6	25	15	15
8	8	8	26	15	20
9	8	10	27	16	16
10	8	12	28	16	20
11	9	7	29	18	18
12	9	9	30	20	20
13	9	14	31	20	24
14	10	10	32	24	24
15	10	12	33	24	30
16	10	14	34	25	25
17	12	12	35	30	30
18	12	14	36	35	35

3.10.1 Lubrication chart or plate. A lubrication chart or plate shall be securely attached to each manipulator. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. A nameplate shall be securely attached to each manipulator. The nameplate shall contain the information listed below. If the manipulator is a special model, the model designation shall include the model of the basic standard manipulator and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

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Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Lift travel (feet)
Reach travel (feet)
Load capacity (pounds)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.12 Workmanship. Workmanship of the manipulator and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

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4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article inspection and all tests shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The manipulators shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The manipulator shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Performance test The manipulator shall be connected to the proper power source. The specified welding head load shall be attached to the proper location on the boom, in its lowered and retracted position, and the following tests performed.

4.7.2.1 Rotary motion test. The manipulator shall be tested for rotary motion by operating the controls to raise the crosshead and loaded boom 3 feet. The column shall then be rotated 180 degrees in each direction at least three times. The rotary motion shall conform with the requirements of 3.4.1.

4.7.2.2 Lift motion test. The manipulator shall be tested for lift motion by operating the controls to raise the crosshead and loaded boom to the top of lift travel specified in table I. The load shall be lowered and raised at least three times. The precision of different rates and the minimum and maximum rates of lift motion shall conform with the requirements specified in 3.4.4.1.

4.7.2.3 Reach motion test. The manipulator shall be tested for reach motion by operating the controls to extend the loaded boom to the end of reach travel specified in table I. The ram or boom shall be retracted and

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extended at least three times. The precision of different rates and the minimum and maximum rates of reach motion shall conform with the requirements specified in 3.4.4.2.

4.7.2.4 Path accuracy test. The manipulator shall be tested for accuracy of path travel by placing the crosshead at the bottom of lift and the ram or boom at retracted reach. With the specified welding head load in position on the boom, and the tip of a welding rod clamped at welding position, the controls shall be operated to extend the ram or boom to full reach at a common welding speed. The actual path of the welding rod tip shall be compared each 12 inches with an ideal straight path or straight edge. The path precision test shall then be repeated at the top of lift. Deviations shall not exceed the tolerances required to perform quality welds in accordance with 3.4.5.

4.7.3 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.9.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. The manipulators covered by this specification are intended for use in welding shops for the versatile and complex positioning of automatic welding heads in the performance of circumferential and longitudinal seam welds.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquistion documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2 and table I).

c. First article, if required (see 3.1).

d. If manipulator design and construction is required in a specific dimensional system (US or SI), state required system (see 3.2.1).

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e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Painting, if different (see 3.3.5).

h. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.9).

i. Base, if different (see 3.4.1).

j. Welding path accuracy, if different (see 3.4.5).

k. Electrical system and input power, if different (see 3.4.6).

l. Motor enclosure, if different (see 3.4.6.1).

m. Control system, if different (see 3.4.7).

n. Size and capacity, if different (see 3.5).

o. Standard equipment, if different (see 3.7).

p. Specify and fully describe optional equipment, if required (see 3.8).

q. Specify applicable repair parts, if required (see 3.9).

r. First article inspection, if different (see 4.4).

s. Quality conformance inspection, if different (see 4.5).

t. If preservation, packing, and marking in accordance with MIL-W-45562 is required, specify level of preservation and packing (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans and acceptance test reports should be specified on a DD Form 1423, Contrast Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

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6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the manipulator itself. It does not require compliance with those OSHA requirements that concern the manipulator in its operating environment, such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. OSHA limits the total hazard level of these hazards in the environment and does not limit the hazard level of the individual manipulator in the environment. The procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional manipulator requirements that will integrate the new manipulator into its future operating environment. The manipulator shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial manipulator supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required because the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1 (f).

6.5 Subject term (key word) listing.

Circumferential seam welding
Concentric circles
External seam welding
Horizontal seam welding
Internal seam welding
Longitudinal seam welding
Manipulator
Metal inert gas (MIG) welding
Table tilt system
Tungsten inert gas (TIG) welding

6.6 Cross reference. The sizes covered by this specification are identical to those covered in the previous edition of this specification.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Preparing activity:
DLA - IP

(Project 3436-0039)

Review activities:
Navy - AS, OS
Air Force - 84
DLA- GS

User activities:
Navy - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-80173B	MANIPULATORS, AUTOMATIC WELDING HEAD, STATIONARY BASE AND TRAVELING BOOM, ELECTRIC MOTOR DRIVEN

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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